

Personal Learning Journey

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Segment Specific Association Mining and Recommendations

Project context. Our group analysed the UCI Online Retail dataset to identify customer segments and segment-specific product associations for retention, reactivation and cross-selling decisions. This reflection explains my individual contribution, learning and development during the project.

My contribution

My responsibility was to discover product relationships within the five validated customer segments and translate them into practical recommendations. I connected 391,150 merchandise transaction rows and 18,402 invoices to 4,334 assigned customers. For each segment, I built a sparse Boolean invoice-product basket so repeated quantities did not inflate co-occurrence. I used FP-Growth with an adaptive minimum support of the greater of 1% or 20 segment invoices and limited itemsets to three products. After initial rule generation, I applied practical thresholds and created segment summaries, sensitivity results, overlap analysis, two figures, output fingerprints and 40 final recommendations.

What I learned

I learned that support, confidence and lift answer different questions and can be misleading when used alone. Confidence may favour products that are already popular, while lift can become very high for rare combinations. Leverage and absolute support count helped me judge whether a relationship had practical evidence. I therefore required at least 25 supporting invoices, confidence of 0.40, lift of 1.50 and positive leverage. This reduced 3,129 initially retained rules to 2,361 practical rules. I also learned why sparse matrices and FP-Growth are appropriate for baskets with thousands of products and densities below 0.72%.

Challenge and judgement

The most difficult part was balancing discovery with reliability across segments of very different sizes. A single global occurrence threshold could remove nearly all patterns from smaller segments, while a very low support value could create unstable rules. I compared support values of 0.008, 0.010, 0.0125 and 0.015 for every segment, producing 20 sensitivity experiments. Stricter settings retained approximately 24% to 54% of baseline rules depending on the segment. This supported 1% as a balanced starting point, followed by the absolute practical filter. I also selected distinct consequent products so the final list was not dominated by repetitive versions of one product family.

Teamwork and future development

My analysis depended on the customer assignments produced by the clustering member and the product identifiers prepared during preprocessing. I validated the merge as many-to-one and checked customer, invoice and segment counts before mining. I then converted technical rules into different objectives and channels for Champions, Loyal High Value, At Risk Regular, Recent Low Engagement and Hibernating Low Value customers. AI assistance helped with debugging and wording, but I verified outputs through assertions, reload checks, fingerprints and parameter comparisons. In future, I would include profit, inventory and campaign-response data and validate recommendations through A/B tests. I learned that association does not prove causation and that responsible recommendations require both statistical evidence and business constraints.